

Lab Assignment 10: BJT Amplifier circuits

Revision: March 31, 2023

Summary

This lab assignment is the continuation of lab assignment number 9 in the sense that you obtained BJT characteristics in the previous lab and in this lab you will design and study a common emitter amplifier with a given set of specifications. You will also study differential amplifiers with a common emitter resistor.

Learning Outcomes: After completing this lab, you should be able to:

- Design and simulate a common emitter amplifier with a given set of specifications.
- Build and design a differential BJT amplifier with an emitter resistor to bias it.
- Do differential and common mode gain measurements.

Required Equipment

- EE352 analog parts kit
- Breadboard
- Function generator, oscilloscope
- DMM
- DC power supplies

I. Single Stage Common Emitter Amplifier

In this part of the lab, you will design a common emitter BJT amplifier as shown in Figure 1 using a 2N3904 npn transistor from your parts kit. The design specifications are:

- Load Resistance, R_L : $10K\Omega$
- Input Resistance, $R_i > 1.2K\Omega$
- Mid-band voltage gain, $G_v = V_o/V_s$: $-85 \pm 20\%$
- Power Supply Voltage: $V_{CC} = +12$ Volts
- Coupling and bypass capacitors: $C_B = 10\mu F$; $C_C = 10\mu F$ and $C_E = 47\mu F$

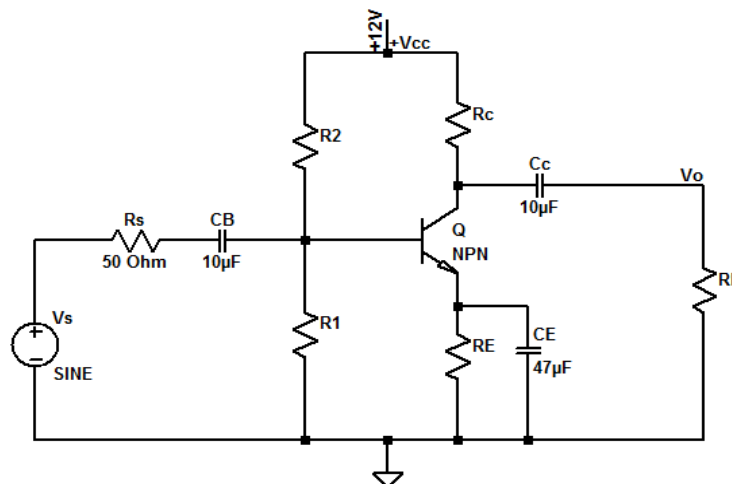


Figure 1. Single stage Common Emitter Amplifier

Pre-lab:

- Begin the design by choosing an appropriate value for the bias current I_E to meet the gain, input and output resistance requirements. Assume $\beta = 100$ and $V_A = 100$ V ($\lambda = 0.01$ V⁻¹). The input resistance is given as:

$$R_i = R_1 \parallel R_2 \parallel r_\pi \quad (1)$$

where

$$r_\pi = V_T / I_B = V_T (\beta + 1) / I_E \quad (2)$$

and the small signal mid-band output resistance at the collector is:

$$R_o = R_C \parallel R_L \parallel r_o \quad (3)$$

where

$$r_o = V_A / I_C$$

and voltage gain is:

$$G_v = -g_m R_o \quad (4)$$

and

$$g_m = I_C / V_T \quad (5)$$

For a design one may consider:

- $R_C \cdot I_C = V_{CC}/2$, for maximum symmetrical swing at the output.
- Collector and emitter currents are approximately equal and, i.e. β = large
- r_o to be very large, i.e. V_A is large.

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(iv) current through R_1 and R_2 is $\sim I_E/10$ and

(v) for DC biasing base voltage is $\sim 1/3 V_{CC}$

Based on these equations and considerations calculate the values for R_1 , R_2 , R_E and R_C .

(b) Simulate the circuit given in Figure 1, using $\beta=h_{fe}=100$ $V_A=100$ V ($\lambda=0.01$ V⁻¹). You may use PSPICE, LTSPICE or ORCAD simulation tools. The simulation should include DC operating voltages and currents; and an AC frequency response from 10Hz to 10MegHz. In your simulation, adjust the values of R_E and R_C until the gain is -85 V/V.

Lab Procedures:

1. Measure all resistors with your DMM and record their values in your lab notebook. Use the RLC meter to measure the capacitance values.
2. With no AC signal applied to the circuit, connect the DC supply voltage as shown in Figure 1. Using DMM, measure and record all DC node voltages; and collector current in the circuit. At this point, your DC measurements should be very close to the expected DC values given by the simulation. Then, estimate the value of g_m from your DC measurements.
3. Apply a 20 mV peak-to-peak, 20KHz sinusoidal input signal at the input. Measure the output voltage with your oscilloscope. Compute the mid-band voltage gain. If the gain is not within the specs of -85V/V $\pm 20\%$, then adjust R_E and /or R_C as needed.
4. *Demonstrate your circuit to the TA and have him or her initial your lab checklist.*
5. Determine the low cutoff frequency (f_L) and the high cutoff frequency (f_H) of the amplifier. To determine f_L , measure V_o at the midband (20 kHz), then decrease the frequency until the output voltage is 0.707 ($1/\sqrt{2}$) of V_o at the midband. This is the low cutoff frequency; record this frequency in your notebook. To determine f_H , increase the frequency until the output voltage is 0.707 ($1/\sqrt{2}$) of V_o at the midband. This is the high cutoff frequency; record this frequency in your notebook.
6. Plot the frequency response of your amplifier (i.e., plot the gain in dB vs. $\log_{10}$ of the frequency). Start from two decades below f_L up to two decades after f_H . Make sure to have enough data points around f_L and f_H to capture the curve smoothly. It is recommended that you plot the data on Matlab or MS Excel as you collect the data. Change x-axis to be in log. Include the frequency response plot in your lab report.
7. *Demonstrate your frequency response plot to the TA and have him or her initial your lab checklist.*
8. Set the frequency to be 20KHz sinusoidal signal and increase its input voltage level till the output, top and bottom start flattening. Measure and record this output voltage. It is the maximum output swing.

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- Build the circuit as shown in Figure 2 for the measurement of input impedance. Reduce the input signal so that there is no distortion at the output. Keeping input signal frequency to be 20KHz measure V1 and V2 and calculate input impedance as $V_2/((V_1-V_2)/1K\Omega)$.

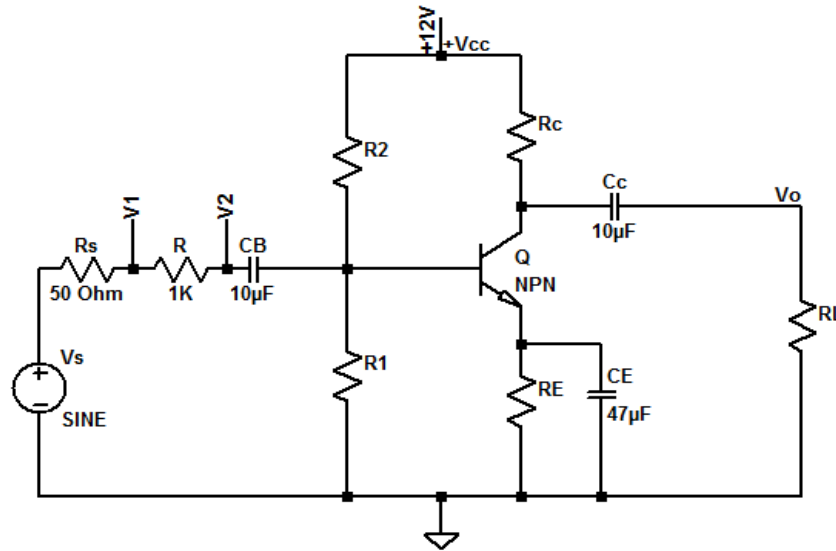


Figure 2. Input Impedance measurement of BJT amplifier

- Divide R_E into $R_{E1} = 47\Omega$ and $R_{E2} = R_E - 47\Omega$ as shown in Figure 3. Calculate and measure new mid-band voltage gain and input resistance. *Demonstrate your results to the TA.*

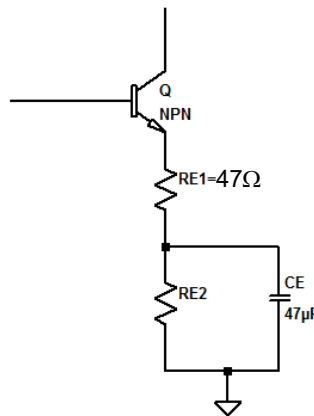


Figure 3. Un-bypassed Emitter Resistance Circuit of a BJT Common Emitter Amplifier

II. Differential Amplifier

In this part of the lab, you will build a basic differential amplifier and measure differential mode (A_d) and common mode (A_c) voltage gains at 1KHz. You will use two 2N3904 NPN transistors to build the differential amplifier. Ideally these transistors should be perfectly matched (same β , r_{π} , r_o , V_A , etc.), but here there will be some mismatch between the transistor parameters.

Pre-Lab: None.

Lab Procedures:

1. Build the basic differential amplifier circuit as shown in Figure 4 using 2N3904 NPN transistors for Q1 and Q2.

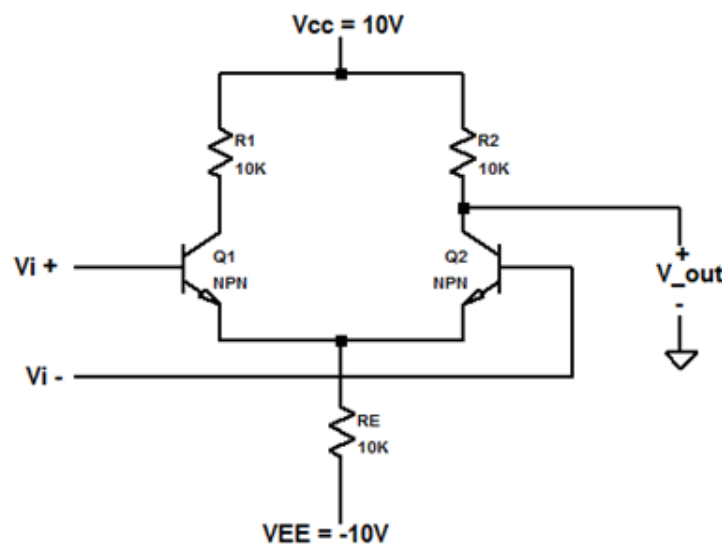


Figure 4. Basic Differential Amplifier

2. Common mode voltage gain is measured by using the Figure 5, where the box represents the basic amplifier of figure 4. Connect 1KHz, 1V peak to peak sinusoidal signal to both + and - inputs. Check the output voltage V_o on a scope for no distortion (AC coupling). Record the input and output voltages and calculate common mode gain $A_c = (V_o/V_{in})$

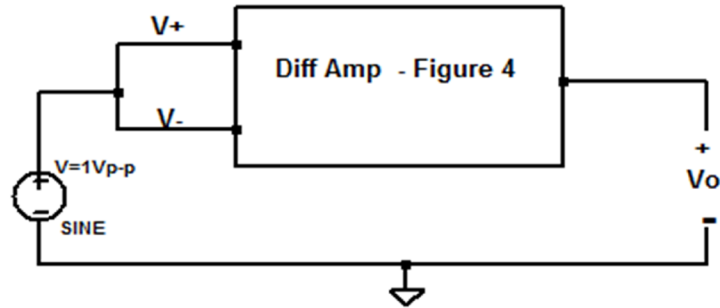


Figure 5 Measurement set-up for common mode gain measurement

3. Differential mode voltage gain A_d is measured using Figure 6, where - input terminal is connected to ground and a small, $\sim 10\text{mV}$ peak to peak sinusoidal signal is connected to + input terminal. Make sure that there is no distortion at the output. Measure input and output voltages (AC coupling) and then calculate differential mode as $A_d = V_o/V_{in}$
4. Compute common mode rejection ratio (CMRR) in dB as $20\log_{10}(A_d/A_C)$ and *demonstrate the operation of the circuit to the TA.*

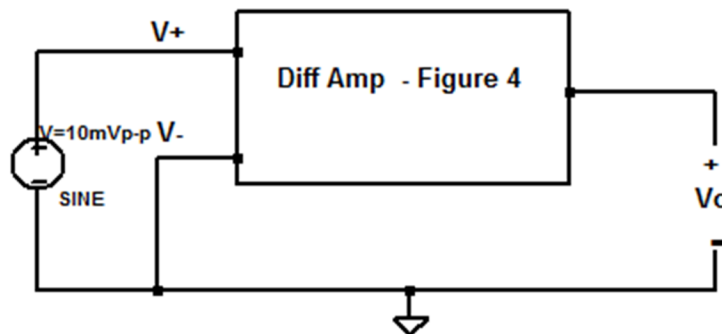


Figure 6. Measurement set-up for Differential Mode voltage gain

Lab Report:

In your lab report, provide a summary of the results of this lab assignment. You should include, at a minimum, all items indicated on the lab checklist. Append the lab checklist sheet with Laboratory Instructor's initials indicating completed lab demos to your report.

Lab 10 Checklist (100 total points)

Name: _____

Lab title and introduction

(5 pts)

- Lab title, your name, date, and lab partner.
- Brief introduction (two or three sentences) explaining the purpose of this lab.

I. Single Stage Common Emitter Amplifier (60 pts total)

1. Diagram of amplifier. (3 pts)
2. Measured component values: R_1 , R_2 , R_E , R_C , R_L , C_E , C_C , C_B (3 pts)
3. Measured DC voltages/currents: V_B , V_E , V_C , and I_E (5 pts)
4. Measured mid-band gain. (5 pts)
- _____ 5. **DEMO**: Have a TA initial this sheet, indicating mid-band operation of the amplifier (6 pts)
6. Plot of measured gain vs. frequency (Bode Plot) (6 pts)
- _____ 7. **DEMO**: Have a TA initial this sheet, indicating that he/she has observed your circuit's frequency response operation. (6 pts)
8. Maximum peak-to-peak output swing. (3 pts)
- 8 Measured input Impedance. (6 pts)
- 9 Mid-band gain and Input impedance when emitter resistance is un-bypassed (6 pts)
- _____ 10. **DEMO**: Have a TA initial this sheet, indicating that he/she has observed your circuit's mid-band gain and input impedance with un-bypassed emitter resistance. (6 pts)
11. Discussion of PSpice / LTSpice gain simulation results and the measured values (5 pts)

II. Differential Amplifier (35 pts total)

1. Diagram of the basic differential amplifier. (7 pts)
2. Measured common mode gain (7 pts)
3. Measured differential mode gain (7 pts)
4. Calculated CMRR (7 pts)
- _____ 5. **DEMO**: Have a TA initial this sheet, indicating that he/she has observed your circuit's operation. (7 pts)